Bayesian Persuasion with Unknown Types of Receivers: An Experimental Investigation

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July 9, 2026

Abstract

Canonical models of Bayesian persuasion assume that the sender knows the receiver's preferences when designing information. In many real-world settings, the sender knows only how common each type of preference is, not which type of receiver she will face. We study Bayesian persuasion with such type uncertainty in a laboratory experiment. By manipulating the receiver's payoff structure, we create two types: one is harder to persuade, while the other is easier to persuade because its payoff makes the sender-preferred action more attractive. Across two between-subjects treatments, we vary only the receiver-type distribution, so that either the hard-to-persuade type or the easier-to-persuade type is more common. We find that senders adjust their plans in the direction predicted by theory: when the easier-to-persuade type is more common, they choose less informative plans. The adjustment, however, is smaller than the theoretical benchmark predicts. We argue that this smaller magnitude reflects a rational response rather than a mere underreaction: the easier-to-persuade type turns out harder to persuade than theory predicts, and senders appear both to anticipate this in their beliefs and to react by supplying more information when that type is more common.

Keywords: Bayesian Persuasion; Experiments; Information Design; Signals.

JEL Codes: C72; C92; D82.

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1 Introduction

Consider a seller who wants a customer to buy her product. To persuade the buyer of its quality, the seller can commit to a test and report its outcome. A strict test makes a favorable result more convincing, but fewer products pass it. If the seller knew which kind of buyer she faced, the logic would be straightforward: she could use a relatively easy test for a buyer who is already inclined to buy, but would need a stricter test for a skeptical buyer who requires stronger evidence. In practice, however, the seller must commit to the test before she knows which kind of buyer will arrive, and can rely only on her knowledge of how common eager and skeptical buyers are in the market.

This type of uncertainty is common in information design and Bayesian persuasion problems. A prosecutor may design an investigation without knowing how much evidence a jury will require for conviction. A school may choose how finely to grade its students without knowing how demanding employers are in hiring decisions. An advertiser, lobbyist, or policymaker may commit to a communication strategy without knowing whether the audience is already receptive or difficult to convince. In all these cases, the sender faces not only uncertainty about the state of the world, but also uncertainty about the receiver's exact preferences. The central question is therefore how a sender should design information when the receiver can be one of multiple possible types.

The canonical Bayesian persuasion framework studies the strategic interaction between a sender (she) and a receiver (he): the sender commits to a plan that produces a message conditional on an unknown state, and the receiver reacts accordingly after observing the message (Kamenica & Gentzkow, 2011). Several experiments have tested this classic framework in the laboratory (e.g., Fréchette et al., 2022; Au et al., 2023; Zhou, 2023; Rentschler & Samad, 2025). However, these studies usually give the sender exact knowledge of the receiver's payoffs, and hence of his type. Motivated by the examples above, we use a theory-based laboratory experiment to study Bayesian persuasion when the sender does not observe the receiver's type and instead knows only the distribution of possible receiver types.

We first develop a simple Bayesian persuasion framework with two types of receiver. The sender commits to a communication plan before the state is realized, knowing only the probability of facing each type. The receiver observes the plan and the message it generates, then chooses an action. The sender wants to induce one particular action regardless of the state, while the receiver wants his action to match the state. The two types of receivers, Type I and Type II, differ in how much they value matching the sender-preferred state. A Type I receiver earns the same payoff from a correct match in either state. A Type II receiver also wants to match the state but earns more when the correct action is the one preferred by the sender. This payoff asymmetry tilts Type II toward the sender-preferred action and makes him easier to persuade than Type I.

Our framework yields predictions about how the sender should design information as the distribution of receiver type changes. An optimal plan always makes the favorable message just strong enough to persuade the harder-to-persuade (Type I) receiver. The main tradeoff concerns the unfavorable message. When Type II is relatively rare, the sender optimally leaves the unfavorable message too weak to persuade either type, so the favorable message is sent as often as possible. With the two messages well separated, this plan is relatively informative. When Type II is sufficiently common, the sender instead strengthens the unfavorable message just enough to persuade Type II. Though this plan sends the favorable message less often, it leads Type II to take the sender-preferred action regardless of which message he receives. This second plan is less informative, because the two messages are closer in the beliefs they induce. Thus, we predict more (resp. less) informative communication when Type I (resp. Type II) is more common.

We then conduct a controlled laboratory experiment to test our theoretical predictions. The experiment utilizes abstract framing and has two between-subjects treatments. Both treatments keep the sender uncertain about the type she faces. In each treatment, receivers privately learn their own type, while senders learn only the probability of facing each type. The treatments differ only in this probability: Type I receivers are common in the unknown-

high treatment but rare in the unknown-low treatment.

The experiment has two parts, each with twenty rounds: a persuasion game as in the model and following the main features of Fréchette et al. (2022), and an individual task in the spirit of Au et al. (2023). In the persuasion game, senders commit to a communication plan, and receivers, after observing both their type and the sender's plan, choose an action for each possible message. Unlike the persuasion game, the individual task is completed by senders and receivers on their own. In each round, a communication plan is drawn from a past round of the persuasion game played by other participants. In the sender's individual task, she observes the plan and the receiver's type, and then predicts how that receiver responded to the plan. This task generates an incentivized measure of the sender's beliefs about receiver behavior. The receiver's individual task mirrors his task in the persuasion game, with one key difference: his choice no longer affects the sender's payoff. Comparing receiver choices across the two parts therefore allows us to examine whether behavior changes when the receiver's decision affects a live sender.

Overall, our experimental results are in line with our main predictions. Senders adjust their plans in the predicted direction: the plans are less informative when a Type II receiver is most likely, and more informative when a Type I receiver is most likely instead. Such directional adjustment across treatments is warranted only if the two types truly differ in how hard they are to persuade. The receiver side confirms that Type II receivers are indeed easier to persuade than Type I. Because the persuasion mechanism operates through probabilistic reasoning, these patterns should be sharper among subjects who reason better about probabilities. We find exactly this: both the treatment difference in sender informativeness and the type difference in how hard receivers are to persuade are stronger among more numerate subjects.

Although senders adjust their plans in the predicted direction, the size of this adjustment is smaller than what the theoretical benchmark predicts. We find that this gap mainly arises from an asymmetric deviation in the treatment where Type II receivers are most likely:

senders supply more information than theory prescribes. This pattern is actually a rational response to how receivers behave. Type I receivers are about as hard to persuade as theory implies, whereas Type II receivers, though still easier to move than Type I, are much harder to persuade than the benchmark implies. As a result, more information is needed to persuade when the receiver is more likely to be Type II. Senders' elicited beliefs suggest that this over-provision is intentional: they indeed perceive Type II receivers as harder to persuade than theory predicts. We also identify a possible channel behind these receiver departures: receivers are directionally harder to persuade when their choices affect a real sender's payoff than when their choices affect only their own.

Our findings speak to the role of receiver-side uncertainty in persuasion and information design. We show that the central comparative-static logic of Bayesian persuasion extends beyond the standard known-receiver environment: when the receiver is harder (resp. easier) to persuade, senders provide more (resp. less) information. Meanwhile, the results show why receiver-type uncertainty is behaviorally important. The sender's problem is not solved simply by replacing the receiver with an average type; it depends on how different receiver types actually respond to information and on what senders believe about those responses. This highlights a channel that has received little empirical attention: private information on the receiver side can shape communication design even when the sender has full commitment over the information structure.[1]

This paper contributes to the growing literature on Bayesian persuasion. Since Kamenica & Gentzkow (2011), a large theoretical literature has studied how a sender can influence a receiver's action by committing to a certain plan (Bergemann & Morris, 2019; Kamenica, 2019). One strand of theories studies the persuasion of a privately informed receiver, whose type the sender cannot observe (Kolotilin et al., 2017; Coen, 2020).[2] A related strand

[1]Meanwhile, there are experimental studies that examine the role of commitment in persuasion games (Fréchette et al., 2022; Zhou, 2023).

[2]As extensions to Kamenica & Gentzkow (2011), several theories have investigated persuasion with heterogeneity in receivers' utility (Wang, 2013; Alonso & Câmara, 2016; Bardhi & Guo, 2018; Chan et al., 2019; Arieli & Babichenko, 2019; Gitmez & Molavi, 2022; Rodríguez Olivera, 2024) or prior beliefs (Kosterina, 2022; de Clippel & Zhang, 2022; Ederer & Min, 2022).

asks how the sender should design information when she is uncertain about the receiver's preferences altogether (Babichenko et al., 2022; Parakhonyak & Sobolev, 2025). Existing experiments on Bayesian persuasion remain limited and focus on whether senders succeed in designing optimal signals (Fréchette et al., 2022; Au et al., 2023), or how the option to fact-check messages affects persuasion (Rentschler & Samad, 2025). Yet a question of broad practical relevance remains unexplored: how a sender persuades when she knows only the distribution of receiver types, not which one she faces. We provide the first experimental evidence on Bayesian persuasion with receiver-type uncertainty by testing whether the central comparative static of the persuasion model survives in this environment.

The paper also relates to a broader literature on strategic communication and information design. Some theory-based experiments compare communication regimes, asking how outcomes change with the sender's mode of communication (Aristidou et al., 2019; Fréchette et al., 2022; Ziegler, 2023). Others examine how information design interacts with a particular feature of the sender's private information, such as the sender's commitment (Zhou, 2023), competition between senders (Wu & Ye, 2023), moral hazard (Du & Shahriar, 2024), trust (Geng & Guan, 2023), and uncertainty about the sender's own objective (Guo & Liu, 2026). In our case, the private information belongs to the receiver. This shift matters because receiver types may each deviate from theory in different ways, producing a sender–receiver dynamic quite different from the single, known-receiver case.

The remainder of the paper is structured as follows. Section 2 presents our theoretical framework. Section 3 describes the experimental design and testable hypotheses. Section 4 shows our experimental results. Section 5 discusses potential underlying mechanisms. Section 6 concludes.

2 Theoretical Framework

We consider a Bayesian persuasion game between a risk-neutral sender and a risk-neutral receiver. The state of the world is binary, $\omega \in \{\text{red}, \text{blue}\}$, with a common prior $\Pr(\omega = \text{red}) = \mu_0$. Before the state is realized, the sender commits to a signal structure (p_r, p_b), where p_r is the probability of sending the message $m = \text{red}$ when the state is red, and p_b the probability of sending $m = \text{red}$ when the state is blue. Once the state is drawn, a message is generated according to this structure, and the receiver, observing both the structure and the realized message, chooses an action $a \in \{\text{red}, \text{blue}\}$. The sender earns 1 whenever the receiver chooses red and 0 otherwise, regardless of the state, so she designs her structure to maximize the probability that the receiver chooses red.

The receiver, instead, wants his action to match the state. We introduce two types of receiver, $T \in \{I, II\}$, who differ in how much they value matching each state. A Type I receiver earns 1 whenever his action matches the state, regardless of which state occurs. A Type II receiver earns the same for matching the blue state but earns $a > 1$ for matching the red state. Given a belief that the state is red with probability μ, a receiver chooses red only when doing so yields a weakly higher expected payoff than choosing blue.[3] Denote τ_T as the lowest probability of the red state at which type T chooses red. The two types of receivers then have $\tau_I = \frac{1}{2}$ and $\tau_{II} = \frac{1}{1+a} < \tau_I$. The additional payoff from a red match lowers the probability of the red state a Type II receiver needs before choosing red, so Type II is easier to persuade than Type I.

Because the sender commits to the signal structure in advance, the receiver can trust each message and update his belief to the posterior it implies. Given a signal structure (p_r, p_b) and prior μ_0, and letting μ_R and μ_B denote the posterior on the red state after the red message and after the blue message, we have:

$$\mu_R = \frac{\mu_0 p_r}{\mu_0 p_r + (1-\mu_0) p_b}, \qquad \mu_B = \frac{\mu_0 (1-p_r)}{\mu_0 (1-p_r) + (1-\mu_0)(1-p_b)}.$$

[3] Type I therefore chooses red when $\mu \geq 1 - \mu$, while Type II does so when $a\mu \geq 1 - \mu$.

Without loss of generality, we label the red message as the one inducing the higher posterior on the red state, so that $\mu_R \geq \mu_0 \geq \mu_B$. We therefore call the red message the stronger message and the blue message the weaker message. Because the two posteriors must average back to the prior, the sender cannot set them freely: she has to trade off which posteriors to induce against how often each occurs.

Were the sender to know the receiver's type, she would face a standard persuasion problem. However, in the current setup she does not observe the type and knows only that the receiver is Type I with probability c_I and Type II with probability $c_{II} = 1 - c_I$. Given an induced posterior μ, the probability for a receiver of unknown type to choose red is:

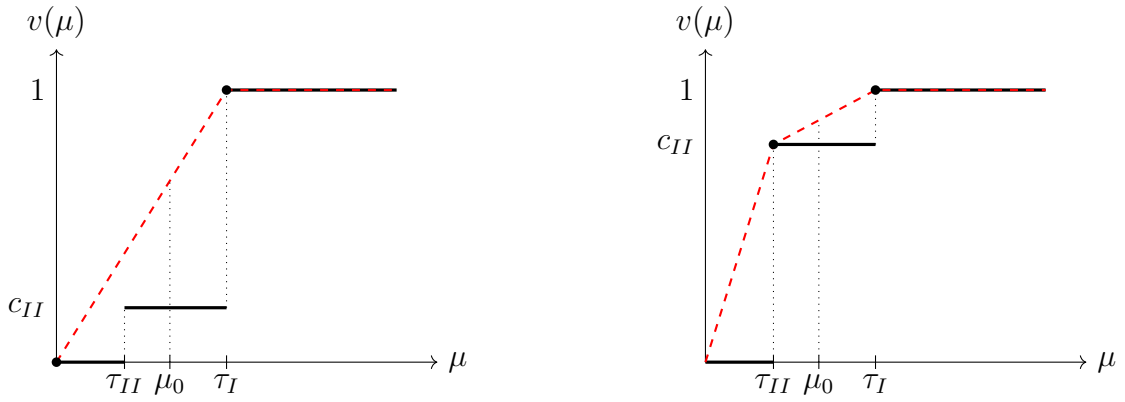
$$v(\mu) = c_I \, \mathbb{1}\{\mu \geq \tau_I\} + c_{II} \, \mathbb{1}\{\mu \geq \tau_{II}\}.$$

Because the sender is paid exactly when the receiver chooses red, $v(\mu)$ is also her payoff at posterior μ. Since the sender designs a signal structure that induces a distribution of two posteriors with mean μ_0, her expected payoff is the average of v over that distribution. The highest achievable payoff for the sender is therefore the concave closure of v evaluated at μ_0 (Kamenica & Gentzkow, 2011).

We focus on the intermediate-prior region $\tau_{II} < \mu_0 < \tau_I$ and leave the two boundary cases, where the prior lies outside this range, to Appendix A. In this region, depending on how common Type II is, the concave closure of v either rises in a single straight segment from 0 to τ_I or bends upward at τ_{II} first (Figure 1), which then determines the sender's optimal structure. We state this formally in the following proposition.

Proposition 1. *Suppose $\tau_{II} < \mu_0 < \tau_I$. The sender's optimal signal structure sets the stronger message to induce $\mu_R = \tau_I$ and the weaker message to induce*

$$\mu_B = \begin{cases} 0 & \text{if } c_{II} \leq \tau_{II}/\tau_I \quad (\textit{Plan A}), \\ \tau_{II} & \text{if } c_{II} \geq \tau_{II}/\tau_I \quad (\textit{Plan B}). \end{cases}$$



(a) $c_{II} < \tau_{II}/\tau_I$: Plan A

(b) $c_{II} > \tau_{II}/\tau_I$: Plan B

Figure 1: The sender's value $v(\mu)$ (solid) and its concave closure (dashed) as a function of the induced posterior on the red state. Filled dots mark the two posteriors the optimal structure induces, and the prior μ_0 lies between them. When Type II is rare (panel a) the kink at τ_{II} lies below the chord and the optimal structure pushes the weaker posterior down to 0 (Plan A). When Type II is common (panel b) the kink rises above the chord and the optimal structure brings the weaker posterior up to τ_{II} (Plan B).

Corollary 1. *Equivalently, the optimal structure in Proposition 1 is*

$$(p_r, p_b) = \begin{cases} \left(1,\ \dfrac{\mu_0(1-\tau_I)}{(1-\mu_0)\tau_I}\right) & \text{under Plan A,} \\ \left(\dfrac{\tau_I(\mu_0-\tau_{II})}{\mu_0(\tau_I-\tau_{II})},\ \dfrac{(1-\tau_I)(\mu_0-\tau_{II})}{(1-\mu_0)(\tau_I-\tau_{II})}\right) & \text{under Plan B.} \end{cases}$$

The two optimal structures (Plan A and Plan B) both set the stronger message just strong enough to persuade Type I and differ only in the weaker message. Under Plan A the weaker message persuades no one, so both types choose blue after the weaker message and red after the stronger one. This plan maximizes how often the stronger message is sent. Under Plan B the weaker message is strengthened just enough to bring in Type II, so Type II chooses red regardless of the message received while Type I still acts only on the stronger one. Compared to Plan A, Plan B sends the stronger message less often but secures Type II unconditionally. Intuitively, this trade becomes worthwhile once Type II is common enough ($c_{II} > \tau_{II}/\tau_I$). Because Plan A's weaker message induces a lower posterior than Plan B's (0 rather than τ_{II}), Plan A spreads its two posteriors farther apart and conveys more

information to the receiver than Plan B.

3 Experimental Design and Testable Hypotheses

3.1 Design and Procedures

To test these predictions, we conducted a controlled laboratory experiment. Each subject was randomly assigned to one of two roles, sender or receiver, and kept this role throughout the experiment. The experiment had two parts, with forty rounds in total. Part 1 (rounds 1-20) involved strategic interaction between a sender and a receiver, while Part 2 (rounds 21-40) consisted of individual tasks without direct interaction.

In Part 1, a sender and a receiver interacted as in our theoretical framework. In each round, senders and receivers were randomly rematched, and each receiver was randomly assigned a type according to the treatment-specific probabilities. The state was the color of a ball drawn from a bag of one red and two blue balls, so it was red with prior probability $\mu_0 = \frac{1}{3}$. Knowing only the probability of facing each type, the sender chose a communication plan (p_r, p_b): she specified the probability of sending the message "the ball is red" in each state, p_r when the drawn ball is red and p_b when it is blue. The message "the ball is blue" was sent with the remaining probability in each state. Figure 2 shows a sender's decision screen in Part 1.

The receiver observed both his assigned type for the round and the sender's communication plan. He then chose a guessing plan, specifying which action, red or blue, he would take for each possible message. Figure 3 shows a receiver's decision screen in Part 1.

After both plans were submitted, the computer drew the ball, sent a message according to the sender's communication plan, implemented the receiver's guess according to the guessing plan, and determined payoffs. The sender earned \$0.40 if the receiver guessed red and \$0 if the receiver guessed blue. A Type I receiver earned \$0.40 for any correct guess and \$0 for an incorrect guess. A Type II receiver earned \$0.40 if the ball was red and the receiver

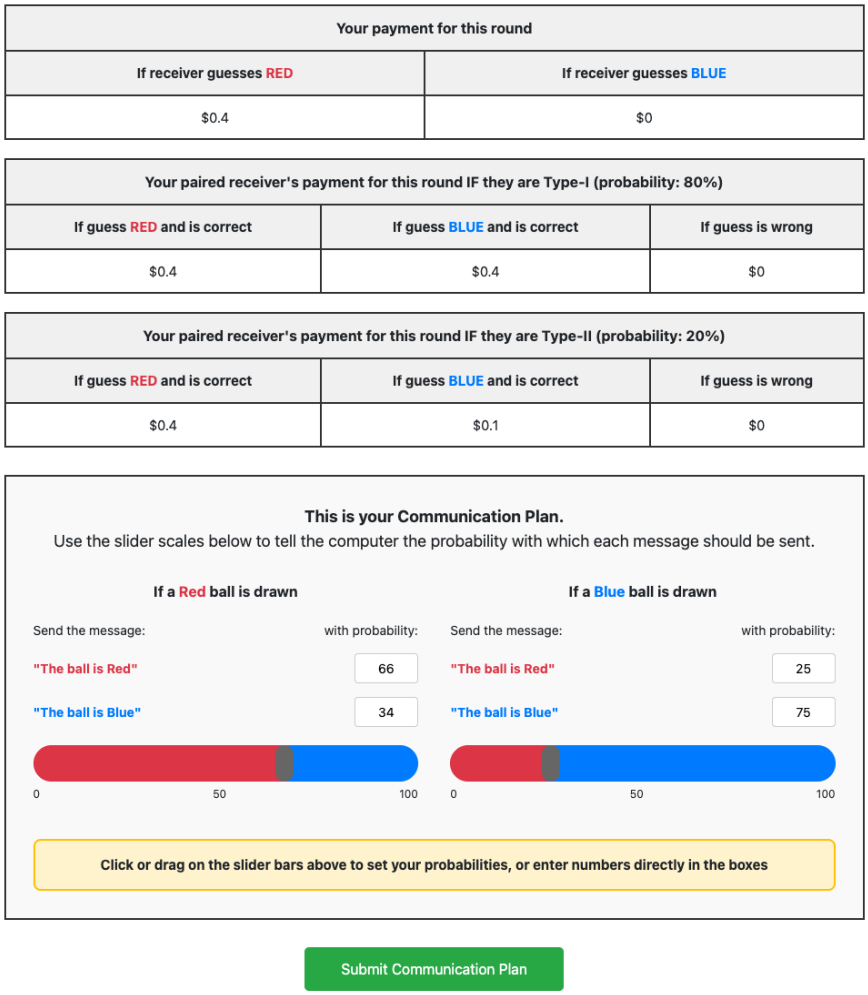


Figure 2: Sample interface of a sender's decision in Part 1.

correctly guessed red, earned \$0.10 if the ball was blue and the receiver correctly guessed blue, and earned \$0 for an incorrect guess. This payoff structure leaves the sender's and receiver's interests only partly aligned and sets the two types apart, as in our framework.

The experiment had two between-subjects treatments that differed only in how likely the matched receiver was to be each type. In the unknown-high treatment (UH), the receiver was Type I with probability 0.8, so $c_{II} = 0.2$. In the unknown-low treatment (UL), the receiver was Type I with probability 0.2, so $c_{II} = 0.8$. These two probabilities provide sufficient contrast that the theory predicts different optimal plans under different treatments.

In Part 2, senders and receivers each completed an individual task.[4] In both tasks, each

[4]Sample interfaces in Part 2 are available in Appendix C.



Figure 3: Sample interface of a receiver's decision in Part 1.

round randomly drew a past round from Part 1 that was played by another pair. The sender was shown the communication plan chosen by another sender in that past round and the type of the matched receiver, and then guessed that receiver's guessing plan. A sender earned \$0.40 if the predicted guessing plan and the actual guessing plan produced the same guess for the realized message, and zero otherwise.

Recall that our theoretical framework assumes that the receiver is a risk-neutral, self-interested expected-utility maximizer who acts at the posterior threshold predicted by the framework. In practice, receivers may depart from this benchmark. A sender may form her own beliefs about how each type actually behaves. Those beliefs should then shape the plan she chooses, and may thereby shift her plan in each treatment beyond what the framework predicts. The sender's individual task in Part 2 is designed to measure these beliefs.

A receiver in Part 2 was shown the communication plan and the assigned type from that round, and his task was to choose a guessing plan as if he were a receiver of that type. His payoff followed the same rule as in Part 1, but because the communication plan came from

a past round rather than a live sender, his choice affected only his own earnings. Comparing a receiver's choices across Parts 1 and 2 therefore isolates whether his decisions depend on facing a live sender.

Subjects for this experiment were 112 undergraduate students at Texas A&M University, recruited using the econdollars.tamu.edu website, a server based on ORSEE (Greiner, 2015). Eleven sessions took place at the Economic Research Laboratory at Texas A&M University from February to May 2026: 54 subjects were in treatment UL, and 58 subjects were in treatment UH.[5] Each subject participated in only one session. Subjects made decisions using the oTree software (Chen et al., 2016). Each session took less than 100 minutes, including instructions, comprehension quiz questions, and payment procedures. Earnings were the sum of each round's payoff plus a \$10 show-up bonus, for an average payment of about \$18.

3.2 Testable Hypotheses

In this subsection, we propose our testable hypotheses grounded in the theoretical framework and the experimental design. Together, these hypotheses allow us to assess how closely actual behavior aligns with the framework.

Hypothesis 1. *The sender's plan is more informative about the state under treatment UH than under treatment UL.*

Section 2 shows that the sender's optimal plan switches from Plan A to Plan B once $c_{II} > \tau_{II}/\tau_I$, which holds under UL but not under UH. Plan A spreads its two posteriors farther apart than Plan B does, making it a more informative plan. The framework therefore predicts a more informative plan under UH and a less informative plan under UL. Specifically, the theoretical benchmark predicts $(p_r, p_b) = (1, \frac{1}{2})$ under UH and $(p_r, p_b) = (\frac{2}{3}, \frac{1}{3})$ under UL. This echoes a familiar intuition from single-receiver Bayesian persuasion: the easier the

[5] Demographics are relatively balanced across the two treatments: in UL (UH), 54% (47%) of subjects are female and the mean age is 21.0 (20.6) years.

receiver is to persuade, the more red actions the sender can induce with a plan that separates the two posteriors less, leading to a less informative plan.

Hypothesis 2. *A Type II receiver acts red at a lower posterior on the red state than a Type I receiver.*

The directional prediction behind Hypothesis 1 rests on the two types acting on different posteriors. Recall that we use τ_T for the posterior on the red state at which a type-T receiver is just willing to act red. The framework predicts $\tau_{II} < \tau_I$, with experimental benchmarks $\tau_I = \frac{1}{2}$ and $\tau_{II} = \frac{1}{5}$.

4 Results

We organize our results around our testable hypotheses. Sections 4.1 and 4.2 test the framework's core prediction and the premise it rests on: that sender plans are less informative under UL than under UH, and that Type II receivers act at a lower posterior than Type I.

4.1 Senders' Plans and Informativeness

Recall that a sender's plan is captured by the pair (p_r, p_b), where p_i is the probability of sending the red message when the drawn ball is of color i, $i \in \{\text{red}, \text{blue}\}$. Figure 4 plots the round-by-round means of p_r and p_b for each treatment, with the theoretical predictions drawn as dashed horizontals. The gap between p_r and p_b is markedly wider in UH than in UL, with the contrast growing with rounds and stabilizing by the last ten rounds. In UL, p_b rises from 0.568 in the first ten rounds to 0.693 in the last ten rounds (paired Wilcoxon $p = 0.025$), bringing p_r and p_b closer together and shifting plans toward less informative ones. In UH, p_r rises from 0.852 to 0.900 ($p = 0.056$), pushing p_r and p_b further apart and shifting plans toward more informative ones.

Previous persuasion experiments with a single receiver type, including Rentschler & Samad (2025), report little within-session learning. The learning we find here may reflect the

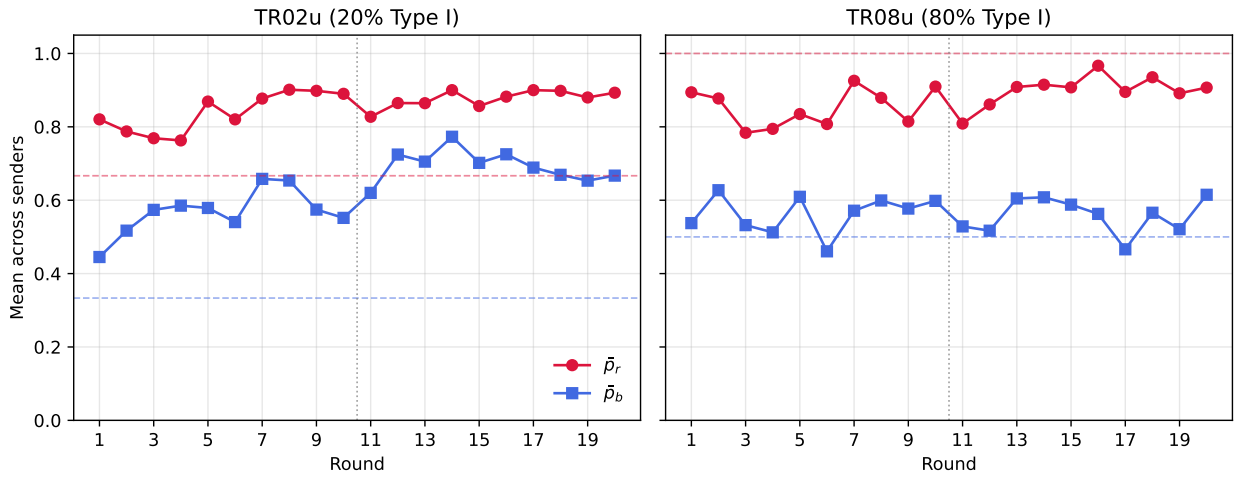


Figure 4: The figure plots mean p_r (red) and p_b (blue) per round, by treatment, where $p_r = P(\text{red message} \mid \text{red ball})$ and $p_b = P(\text{red message} \mid \text{blue ball})$. Dashed horizontals show the theoretical prediction ($p_r = 2/3$, $p_b = 1/3$ in UL; $p_r = 1$, $p_b = 1/2$ in UH).

Table 1: Sender plan summary by treatment

Measure	UL	UH	Δ (UL $-$ UH)	MW 2-s
p_r	0.877	0.900	-0.023	0.598
p_b	0.693	0.558	$+0.135$	0.059
ψ^B	0.164	0.255	-0.091	0.098

Per-sender unit, $n = 27$ (UL) / 29 (UH). $p_r = P(\text{red message} \mid \text{red ball})$ and $p_b = P(\text{red message} \mid \text{blue ball})$. $\psi^B = (p_r - p_b)(\mu_R - \mu_B)$, where μ_R and μ_B are the posteriors on the Red state after the red and blue signals. MW 2-s = Mann-Whitney two-sided.

added complexity of facing an unknown-type receiver. We therefore use the last ten rounds (rounds 11–20) as the primary analysis window for the remainder of this section.

We now compare sender plans across treatments after they stabilize in the last ten rounds. For each measure, we collapse a sender's choices over rounds 11-20 into a single per-sender mean, then compare the two treatments with a two-sided Mann-Whitney test. As shown in Table 1, p_r stays near 1 in both treatments (0.877 in UL, 0.900 in UH), and the cross-treatment difference lives almost entirely in p_b (0.693 vs 0.558, $p = 0.059$). In UL, p_b is higher than in UH, bringing p_r and p_b closer together and making the blue message less informative about the ball's color, consistent with the theoretical prediction.

Since the sender chooses p_r and p_b jointly, comparing them separately may not capture the full picture of plan informativeness. A natural alternative is the correlation between the true ball color and the receiver's guess. However, this measure is comparable across treatments only when the receiver has the same fixed utility. In our case, a cross-treatment comparison of this correlation would conflate changes in the sender's plan with changes in the type composition of receivers.

To summarize plan informativeness in a single number that does not condition on a particular receiver type, we follow Fréchette et al. (2022)'s method and use $\psi^B = (p_r - p_b)(\mu_R - \mu_B)$, which combines the state-dependence of the message, $p_r - p_b$, and the posterior separation it induces, $\mu_R - \mu_B$. Our theory predicts $\psi^B = 0.25$ under UH and 0.10 under UL, with a gap of 0.15. Empirically, ψ^B is higher in UH (0.255) than in UL (0.164), with a gap of 0.091 (Mann–Whitney $p = 0.098$). Thus, sender plans are directionally and marginally less informative in UL than in UH, providing suggestive support for Hypothesis 1.

Result 1. *Consistent with Hypothesis 1, sender plans are less informative under UL than under UH. The difference comes mainly from a higher p_b in UL.*

Although senders adjust their plans in the predicted direction across treatments, each empirical plan deviates from its theoretical counterpart asymmetrically. UH's plan (0.900, 0.558) sits close to the Plan A values $(1, \frac{1}{2})$. UL's plan (0.877, 0.693), by contrast, lies far from the Plan B values $(\frac{2}{3}, \frac{1}{3})$. The same asymmetry appears in ψ^B: empirical informativeness is close to theory under UH (0.255 vs 0.25) but much higher than theory under UL (0.164 vs 0.10). This does not necessarily mean that UL senders behave more irrationally. One possibility is that receivers themselves deviate asymmetrically from theory, and UL senders are responding rationally to how receivers actually behave. As we show on the receiver side in the next subsection, Type II receivers are indeed harder to persuade than $\tau_{II} = 0.2$ suggests, so the asymmetric gap may reflect senders adjusting for this reluctance rather than making an error.

4.2 Receivers Respond to Incentives by Type

The sender's treatment response is warranted only if the two receiver types behave differently, which Hypothesis 2 formalizes as Type II acting red at a lower posterior than Type I. Our design assigns each type a payoff structure intended to produce this gap, and we now test whether receivers respond as intended.

Following Fréchette et al. (2022), we search a grid of candidate posterior cutoffs and assign each receiver the threshold τ^* that best matches his observed (posterior, guess) pairs. Because each receiver plays a Type I receiver in some rounds and a Type II receiver in others, we estimate τ^* separately for each receiver under each type, taking the (receiver $\times$ type) pair as the unit of analysis.

Before comparing thresholds across types, we first ask whether a single threshold is a meaningful way to summarize a receiver's choices at all. A threshold presumes a particular shape of behavior: the receiver is more likely to guess red as the posterior rises, switching from blue to red at a single cutoff. We plot the share of red guesses against the posterior for each type. As reflected in Figure 5, both types' curves tend to rise with the posterior on the red state. This is not merely an aggregate pattern. More importantly, at the individual level, a single threshold fits each receiver's choices well. Averaged across receivers, it reproduces 91% of a Type I receiver's actual guesses and 86% of a Type II receiver's.[6] Only a small minority is not well described by any single threshold: 11% of Type I and 14% of Type II receivers have fewer than 70% of their guesses matched. τ^* therefore provides a reliable summary of how receivers respond.

We now compare τ^* across different types, taking each (receiver $\times$ type) cell as one observation ($n = 54$ for Type I and $n = 51$ for Type II). Pooling both treatments, the average τ^* is 0.52 for Type I and 0.42 for Type II. The difference goes in the predicted direction: Type I receivers require a higher posterior to guess red, and are therefore harder

[6] Some receivers never vary their guess within a type (19% of Type I and 14% of Type II), so a single threshold fits them trivially. Restricting to those who switch, accuracy is still 88% and 84%.

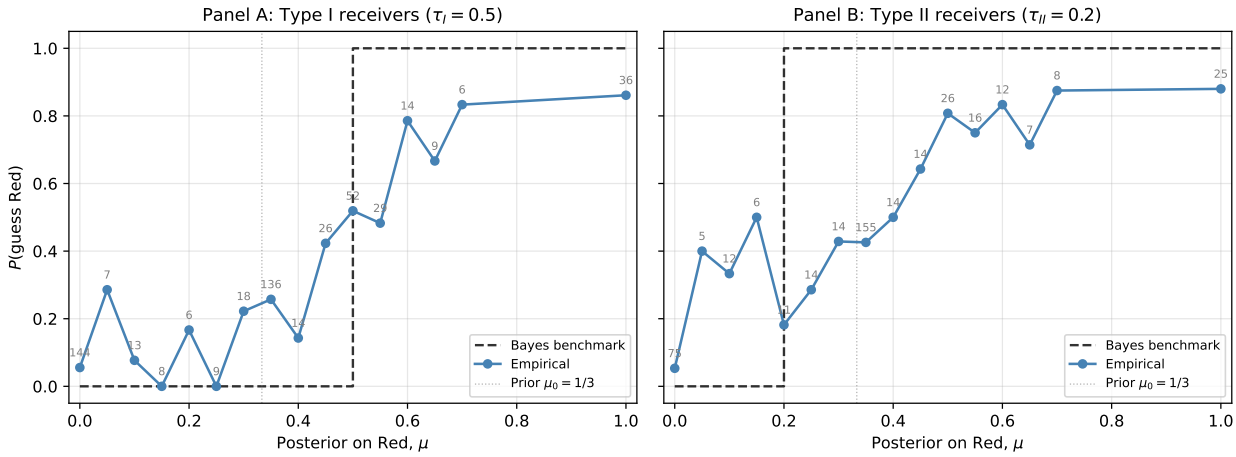


Figure 5: Empirical response curves $P(\text{guess Red} \mid \mu)$ by receiver type, last-10 BP rounds. Each marker is a posterior bin (width 0.05). The number above each marker is the bin's observation count, and bins with fewer than 5 observations are omitted. The dashed step is the predicted receiver action, with the jump at $\tau_I = 0.5$ (Type I) and $\tau_{II} = 0.2$ (Type II). The dotted vertical line marks the prior $\mu_0 = 1/3$.

to persuade (Mann-Whitney $p = 0.026$).[7]

Result 2. *Consistent with Hypothesis 2, Type II receivers act red at a significantly lower posterior than Type I receivers, and are thus easier to persuade.*

Though the direction is as predicted, the two types differ sharply in how closely they match their theoretical thresholds: Type I receivers sit almost exactly at their predicted value (0.52 versus 0.5), while Type II receivers sit well above theirs (0.42 versus 0.2). The predicted gap of 0.30 is thus only 0.10 in the data, with the compression coming entirely from the Type II side. This implies that, in practice, Type II receivers are considerably harder to persuade than the theory predicts. This asymmetry helps interpret the sender-side results in Section 4.1. There, we found that senders facing mostly Type II receivers in UL departed more from their theoretical plan, while senders facing mostly Type I receivers in UH stayed close to theirs. The receiver-side evidence suggests that this departure may reflect a rational response to actual receiver behavior: since Type II receivers are empirically more reluctant

[7]Type I's threshold exceeds Type II's within each treatment, not only in the pool (UH: 0.55 vs 0.40; UL: 0.49 vs 0.44), and the gap is robust to within-receiver pairing: among the 49 receivers with an estimable threshold for both types, a Wilcoxon signed-rank test gives $p = 0.021$.

to guess red than the theoretical threshold $\tau_{II} = 0.2$ implies, senders who are likely to face them should convey more information than the benchmark model prescribes.

5 Discussion of Underlying Mechanisms

The results in Section 4 support our two central predictions: sender plans are less informative when the easier-to-persuade type is more common, and Type II receivers act red at lower posteriors than Type I receivers. However, behavior does not match the theoretical benchmarks exactly. In particular, Type II receivers are substantially harder to persuade than the model predicts, and sender plans in UL are correspondingly more informative than what theory predicts.

In this section, we use three additional measures built into the experiment to examine three mechanisms that may account for these deviations. First, the senders' individual task allows us to measure whether senders anticipate the observed reluctance of Type II receivers. Second, the receivers' individual task allows us to test whether receiver behavior changes when their choices affect a real sender's payoff. Third, the post-experiment probability-numeracy questions allow us to ask whether the main patterns are sharper among subjects who are better able to reason about the posteriors induced by a communication plan.

5.1 Senders' Beliefs about Receiver's Behavior

We have found that senders deviate from the theoretical plan more in the treatment dominated by Type II receivers (UL), and we argued that this asymmetry is rational, since Type II receivers are indeed harder to persuade than the theory predicts. But one may still wonder: do senders hold such a belief and intend to act this way, or did it happen simply by chance?

The individual task gives us a sharper view. In the task, senders are incentivized to guess how a receiver of each type reacts to a plan, which gives a direct measure of their beliefs.

We summarize each belief with a threshold, just as we did for receiver behavior: τ_T^{perc} is the posterior at which the sender expects a Type-T receiver to switch to guessing red. We then ask whether senders' beliefs justify the asymmetry in their plan, that is, whether they see Type II as harder to persuade than theory predicts while viewing Type I in line with it.

According to senders' responses in the individual task, their average τ_T^{perc} is 0.50 for Type I and 0.35 for Type II (Table 2, Panel A). For Type I, what senders believe (0.50) matches what theory predicts (0.50, $p = 0.91$). For Type II, what senders believe (0.35) lies well above what theory predicts (0.20, $p < 0.001$), reflecting that senders indeed judge Type II harder to persuade than the framework predicts. This further supports our argument in Section 4.1: senders deviate more from the theoretical plan by providing more information under UL, where Type II dominates, because they perceive Type II as harder to move, not because they behave less rationally than under UH.

Getting the direction right, however, is not the same as getting the magnitude right. Against how Type II receivers actually behave, senders are too optimistic. Figure 6 plots each type's belief curve against the receivers' actual one. For Type I the two track each other closely. If anything, senders slightly under-predict how often Type I guesses red at the highest posteriors. For Type II the belief curve sits above the actual one across the intermediate posteriors (roughly 0.2 to 0.4): senders expect Type II receivers to be easier to persuade than they actually are. This pattern emerges clearly when we compare the thresholds (Table 2, Panel B). For Type I the perceived and empirical thresholds nearly coincide (0.50 versus 0.52, $p = 0.56$). For Type II the perceived threshold lies well below the empirical one (0.35 versus 0.42, $p = 0.057$). Senders therefore sense correctly that Type II is harder to persuade than theory predicts, but they underrate how much harder: their beliefs stay over-optimistic about moving Type II.

Result 3. *Senders' beliefs offer a plausible explanation for their plan deviations: they perceive Type I receivers almost exactly as theory predicts, but Type II receivers as substantially harder to persuade than theory implies. This asymmetry matches the larger plan deviation*

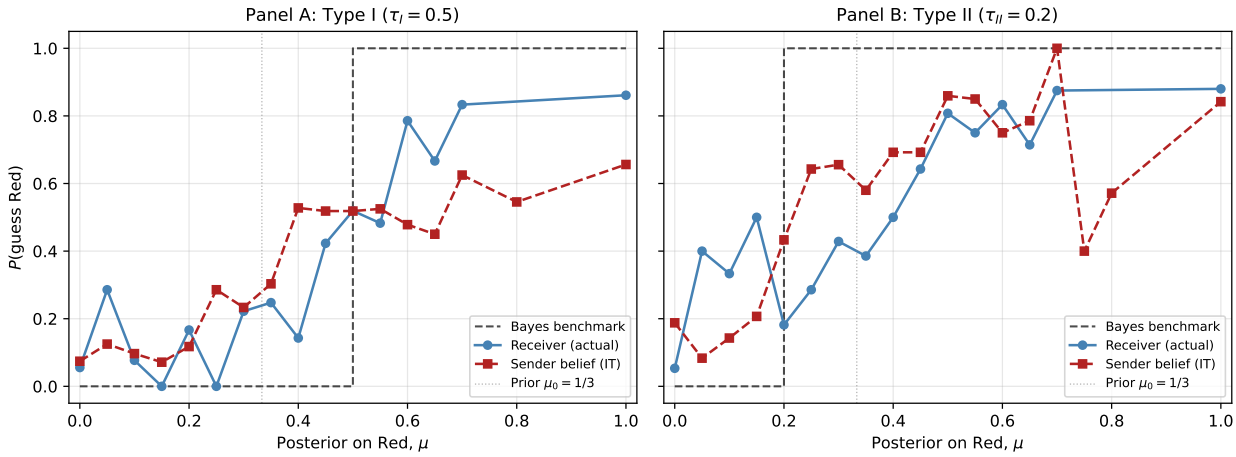


Figure 6: Belief versus behavior: response curves $P(\text{guess Red} \mid \mu)$ by receiver type. The solid blue line (*Receiver, actual*) is receivers' guessing plans over the last-10 BP rounds. The dashed red line (*Sender belief*) is senders' beliefs, across all Individual-Task rounds. Each marker is a posterior bin (width 0.05), and bins with fewer than 5 observations are omitted. The dashed step is the predicted receiver action, with the jump at $\tau_I = 0.5$ (Type I) and $\tau_{II} = 0.2$ (Type II). The dotted vertical line marks the prior $\mu_0 = 1/3$.

in UL, where Type II receivers are most likely.

Taken together, the individual task shows that senders' beliefs are consistent with their plans: both depart from theory along the same asymmetry. For Type I, senders' beliefs match how receivers actually behave, and that behavior is close to theory. For Type II, senders correctly sense that the type is harder to persuade than theory predicts. These beliefs map onto their plans across the two treatments: in UH, where Type I receivers are most likely, senders' plans stay close to the benchmark; in UL, where Type II receivers are most likely, they provide more information than theory calls for. However, senders under-rate how much harder Type II is, still treating it as more persuadable than it really is. With more accurate beliefs about Type II, senders might supply more information under UL, narrowing the cross-treatment informativeness gap documented in Section 4.1 a little further.

Table 2: Senders' perceived thresholds and their calibration against receiver behavior.

Panel A: Perceived threshold versus theory, by type

Type	Belief τ^{perc}	Theory	p
Type I	0.50	0.50	0.91
Type II	0.35	0.20	< 0.001

Panel B: Perceived threshold versus receiver behavior, by type

Type	Belief τ^{perc}	Receiver τ^*	p
Type I	0.50	0.52	0.56
Type II	0.35	0.42	0.055

Per-(sender, type) cell. τ_T^{perc} is the threshold best fitting the sender's predicted Type-T guesses (grid search over $\{0, 0.01, \ldots, 1\}$), pooling all Individual-Task rounds across both Unknown treatments. Panel A: p from a one-sample Wilcoxon signed-rank test of τ^{perc} against the theoretical threshold. Panel B: Receiver τ^* is the empirical threshold from Section 4.2, and p from a Mann-Whitney two-sided test of perceived against empirical thresholds.

5.2 Other-regarding Preferences of Receivers

Note that the receiver can act at a posterior threshold different from the one predicted by theory. Such a departure may stem from other-regarding preferences: he may care about how his action affects the sender's payoff and adjust his behavior accordingly. The receiver's individual task lets us isolate such other-regarding preferences, since it matches the persuasion game except that his guess affects only his own earnings, not a real sender's. If concern for the sender ever enters the receiver's decision, his behavior should differ across the two tasks. We expect the direction to be consistent with what Au et al. (2023) documents: the receiver tends to be harder to persuade once a real sender sits on the other side.

As in Section 4.2, we summarize a receiver's behavior by the posterior threshold above which he guesses red, but here we pool across the two types within a treatment rather than estimating a separate threshold for each. Within a treatment, the type composition and the informativeness of the plans are similar across the two tasks,[8] so the pooled threshold is

[8]Within each treatment the share of Type I rounds is nearly identical across the two tasks (0.22 and 0.22 in UL, 0.84 and 0.82 in UH), and a Kolmogorov-Smirnov test cannot reject equality of the posterior distributions the plans imply ($p = 0.47$ in UL, $p = 0.74$ in UH).

Table 3: Receiver decision threshold in the persuasion game (last ten rounds) versus the individual task, pooled over receiver type. Unknown treatments.

Treatment	τ^*_{BP}	τ^*_{IT}	$\Delta\tau^*$	p
UL	0.428	0.374	+0.055	0.061
UH	0.524	0.480	+0.044	0.148

τ^* is estimated as in Section 4.2 but pooling both types within each treatment. $\Delta\tau^* = \tau^*_{\text{BP}} - \tau^*_{\text{IT}}$, within-subject paired. p from a Wilcoxon signed-rank two-sided test.

comparable between them. Pooling also keeps all the data in play at the individual level,[9] since a type-by-type split would leave the minority type with too few observations in each treatment to estimate its threshold on its own. Although the pooled threshold mixes the two type-specific thresholds and is not a decision rule any real receiver follows, the difference in the thresholds between the two tasks still gives an informative, treatment-level summary of the receiver's other-regarding response.

Table 3 reports the comparison of the pooled threshold between the individual task and the persuasion game within each treatment. In both treatments, the threshold a receiver applies in the persuasion game on average lies above the one he applies in the Individual Task. This suggests he needs a higher posterior before guessing red when his choice also determines a real sender's earnings, by 0.055 in UL (signed-rank $p = 0.061$) and 0.044 in UH (signed-rank $p = 0.148$). Both estimates point the same way: with a real sender on the other side, the receiver tends to be more cautious about acting in the sender's favor.

Result 4. *Receivers are directionally harder to persuade when their choices affect a real sender's payoff. The estimated threshold is higher in the persuasion game than in the individual task in both treatments, although the evidence is statistically modest.*

[9] Such a within-subject contrast is feasible only for the receiver. On the sender side the two tasks differ in both the information available and the nature of the task itself, so the sender cannot be pooled and compared at the individual level.

5.3 The Role of Probability Reasoning

Bayesian persuasion games require probability reasoning from both players: the sender to design the plan and the receiver to read the posterior it implies. If limited probability reasoning is one of the drivers of players' deviations, the theory's predictions should hold more sharply among those who reason about probability well. In the post-experiment survey, we measure each subject's probability numeracy with four questions adapted from Hudomiet et al. (2018), which test understanding of probability along different dimensions. A subject is classified as numerate if she provides correct answers to all questions. 66% of senders and 54% of receivers meet this cutoff.[10]

The results in the persuasion game sharpen substantially within the numerate subgroup.[11] On the sender side, numerate senders indeed make their plans far less informative when the easier-to-persuade Type II is the dominant receiver. Specifically, the magnitude of this reduction is 0.186 in ψ^B (Mann-Whitney $p = 0.019$). The gap roughly doubles the full-sample value of 0.091. On the receiver side, numerate receivers also draw a sharper line between the two types, separating their decision thresholds by 0.188 (0.617 for Type I against 0.429 for Type II, $p = 0.003$), again nearly twice the full-sample gap.

Surprisingly, the receiver's reluctance to be persuaded by a real sender grows stronger among numerate receivers rather than fading. Section 5.2 already showed, for the full sample, that receivers tend to hold a higher threshold in the persuasion game than in the individual task, consistent with Au et al. (2023). One might expect the gap to shrink among the most numerate receivers: a receiver who reads the posterior accurately better understands that acting on it already maximizes his own payoff, so whether a real sender gains from his guess should make less difference to how he responds. Instead, the gap widens in UH, reaching 0.126 among numerate receivers ($p = 0.009$), while in UL it is essentially unchanged at 0.058

[10] Interfaces of the four questions are available in Appendix C. Meeting the cutoff identifies subjects with a better grasp of probability rather than exceptional ability.

[11] The numerate share does not differ significantly between UL and UH for both receivers (56% versus 52%, Fisher exact $p = 0.80$) and senders (56% versus 76%, $p = 0.16$).

($p = 0.064$). These suggest that understanding better what the signal implies does not make a receiver care less about who benefits from his choice. Other-regarding concern thus enters the receiver's utility robustly, rather than washing out as his grasp of the signal sharpens.

Result 5. *The main persuasion patterns are substantially sharper among probability-numerate subjects. Numerate senders reduce plan informativeness more strongly when Type II receivers are common, and numerate receivers display a larger threshold gap between Type I and Type II receivers.*

6 Conclusion

This paper studies Bayesian persuasion when a sender faces uncertainty over receiver types. We develop a simple model in which receivers differ in their payoff incentives and therefore in the posterior threshold required to take the sender-preferred action. The model predicts that the sender should choose a less informative signal structure when she is more likely to face the easier-to-persuade type. We test this prediction in a laboratory experiment in which senders choose communication plans and receivers choose action plans after observing both the plan and the realized message.

Our experimental results are broadly consistent with our main prediction: sender plans are less informative when the easier-to-persuade type is more common. The data reveal asymmetric departures from the benchmark. Though easier-to-persuade type receivers indeed act at a lower posterior than harder-to-persuade type receivers, they are far less persuadable than theory implies. Senders sense this deviation and design more informative plans accordingly, but they are still too optimistic about how hard it is to persuade receivers whose types are easier to be persuaded. We also find suggestive evidence that sender's belief about receiver's behavior, other-regarding preferences, and probability reasoning serve as behavioral mechanisms that explain subjects' behavior.

Our findings point to two different avenues for future work. The first is persuasion under

misspecified beliefs: senders may understand receivers' payoff structures correctly but hold incorrect beliefs about how hard these receivers are to persuade. The second is the role of complexity in Bayesian persuasion and information design, since receiver heterogeneity makes the sender's problem more complex and optimal persuasion depends on the distribution of receiver types and their different responses to signals. Understanding how senders simplify, specify, or adapt to such complexity adds new insights to two different strands of flourishing literature: information design and behavioral game theory.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work, the authors used ChatGPT and Claude to proofread existing content. After using this tool/service, the authors reviewed and edited the content as needed. The authors take full responsibility for the content of the publication.

Funding Sources

This study is supported by the Department of Economics at Texas A&M University, the John Van Hyuck Endowed Fellowship at Texas A&M University, the Fundamental Research Funds at Shandong University (Grant number: 11020064510004), and the Natural Science Foundation of Shandong Province (Grant number: ZR2025QC1251).

Acknowledgments

The earliest version of this study was completed when Jinliang Liu was a Ph.D. student at Texas A&M University. We are indebted to Alexander L. Brown for his extensive discussions and constant support. We thank Catherine Eckel, Huiyi Guo, Silvana Krasteva,

and Zhongheng Qiao for helpful discussions. This study was approved by the Institutional Review Board (IRB) at Texas A&M University (IRB2025-1556M). All errors are our own. Declarations of interest: none.

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Appendices: Not Intended for Publication

A Theory Appendix

A signal structure induces a distribution τ over posteriors, and by Bayes plausibility the average posterior must equal the prior: $\mathbb{E}_\tau[\mu] = \mu_0$. The sender's expected payoff can be expressed as $\mathbb{E}_\tau[v(\mu)]$, where

$$v(\mu) = c_I \, \mathbb{1}\{\mu \geq \tau_I\} + c_{II} \, \mathbb{1}\{\mu \geq \tau_{II}\}, \qquad 0 < \tau_{II} < \tau_I < 1, \quad c_I + c_{II} = 1.$$

Recall that τ_I and τ_{II} are the lowest posteriors on the red state at which a Type I and a Type II receiver, respectively, choose red, with $\tau_{II} < \tau_I$ because Type II is easier to persuade; c_I and c_{II} are the probabilities that the receiver is of each type. The sender therefore solves

$$\max_{\tau} \; \mathbb{E}_\tau[v(\mu)] \quad \text{s.t.} \quad \mathbb{E}_\tau[\mu] = \mu_0.$$

Since both the objective and the constraint are linear in τ, an optimum is attained at an extreme point of the feasible set. That set is defined by two linear equality constraints: the weights of τ sum to one, and its weighted average equals μ_0. An extreme point of such a set is supported on at most two posteriors (Kamenica & Gentzkow, 2011). To characterize the signal structure, we denote the two induced posteriors as $\mu_B \leq \mu_0 \leq \mu_R$, sent with probabilities $1 - q$ and q, respectively. Bayes plausibility $q\mu_R + (1 - q)\mu_B = \mu_0$ determines the probability of the stronger message,

$$q = \frac{\mu_0 - \mu_B}{\mu_R - \mu_B}, \tag{1}$$

and the sender's expected payoff from this two-posterior structure is

$$\Pi = q \, v(\mu_R) + (1 - q) \, v(\mu_B).$$

Proof of Proposition 1. Let $\tau_{II} < \mu_0 < \tau_I$. The stronger message induces the higher posterior $\mu_R \geq \mu_0$, and the weaker message the lower posterior $\mu_B \leq \mu_0$. Since $\mu_R \geq \mu_0 > \tau_{II}$, we have $v(\mu_R) = 1$ if $\mu_R \geq \tau_I$ and $v(\mu_R) = c_{II}$ otherwise. Since $\mu_B \leq \mu_0 < \tau_I$, we have $v(\mu_B) = c_{II}$ if $\mu_B \geq \tau_{II}$ and $v(\mu_B) = 0$ otherwise.

High posterior. Fix the weaker posterior μ_B and consider the choice of μ_R. If $\mu_R < \tau_I$, even the stronger message falls short of the Type I threshold, so $v(\mu_R) = c_{II}$. Since $v(\mu_B) \leq c_{II}$ as well, the sender cannot earn more than $v(\mu_0) = c_{II}$, the payoff from revealing nothing. An informative plan can do better only when the stronger message persuades Type I, i.e. $\mu_R \geq \tau_I$. Among all such μ_R, the smallest one, $\mu_R = \tau_I$, is optimal, since it sends the stronger message as often as possible. Since this holds for every μ_B, the optimal plan either reveals nothing or sets $\mu_R = \tau_I$. We fix $\mu_R = \tau_I$, optimize over the weaker posterior, and verify below that the result beats revealing nothing.

Low posterior. Fix $\mu_R = \tau_I$. If $\mu_B \geq \tau_{II}$ then $v(\mu_B) = c_{II}$ and $\Pi = c_{II} + (1 - c_{II})q$ with $q = (\mu_0 - \mu_B)/(\tau_I - \mu_B)$. Since $\partial q/\partial \mu_B = (\mu_0 - \tau_I)/(\tau_I - \mu_B)^2 < 0$, Π is a decreasing function of μ_B, so the optimal $\mu_B = \tau_{II}$:

$$\Pi_B = c_{II} + (1 - c_{II})\,\frac{\mu_0 - \tau_{II}}{\tau_I - \tau_{II}} \qquad \text{(Plan B)}. \tag{2}$$

If $\mu_B < \tau_{II}$ then $v(\mu_B) = 0$ and $\Pi = q = (\mu_0 - \mu_B)/(\tau_I - \mu_B)$, again a decreasing function of μ_B, so the optimal $\mu_B = 0$:

$$\Pi_A = \frac{\mu_0}{\tau_I} \qquad \text{(Plan A)}. \tag{3}$$

Comparison. Both plans set $\mu_R = \tau_I$ and differ only in μ_B. Rewriting $\Pi_B = [\,c_{II}(\tau_I - \mu_0) + (\mu_0 - \tau_{II})\,]/(\tau_I - \tau_{II})$ and subtracting,

$$\Pi_B - \Pi_A = \frac{c_{II}(\tau_I - \mu_0) + (\mu_0 - \tau_{II})}{\tau_I - \tau_{II}} - \frac{\mu_0}{\tau_I} = \frac{(\tau_I - \mu_0)\,(c_{II}\tau_I - \tau_{II})}{\tau_I(\tau_I - \tau_{II})}. \tag{4}$$

The denominator and $\tau_I - \mu_0$ are positive, so $\Pi_B \geq \Pi_A \iff c_{II}\tau_I \geq \tau_{II} \iff c_{II} \geq \tau_{II}/\tau_I$.

Hence the optimum is Plan A ($\mu_B = 0$) when $c_{II} \leq \tau_{II}/\tau_I$ and Plan B ($\mu_B = \tau_{II}$) when $c_{II} \geq \tau_{II}/\tau_I$, with indifference at equality. The optimal plan between Plan A and Plan B pays at least c_{II}, so it beats revealing nothing, which proves the proposition. □

Proof of Corollary 1. Fix target posteriors $\mu_B \leq \mu_0 \leq \mu_R$ and let $q = \Pr(m = \text{red})$ be the total probability of the stronger message, given by (1). Since $\Pr(m = \text{red}, \omega = \text{red}) = \mu_0 p_r$ and $\mu_R = \mu_0 p_r / q$, we have $\mu_0 p_r = q\mu_R$ and hence

$$p_r = \frac{q\,\mu_R}{\mu_0} = \frac{\mu_R}{\mu_0} \cdot \frac{\mu_0 - \mu_B}{\mu_R - \mu_B}.$$

Likewise $\Pr(m = \text{red}, \omega = \text{blue}) = (1 - \mu_0)p_b = q - \mu_0 p_r = q(1 - \mu_R)$, so

$$p_b = \frac{q\,(1 - \mu_R)}{1 - \mu_0} = \frac{1 - \mu_R}{1 - \mu_0} \cdot \frac{\mu_0 - \mu_B}{\mu_R - \mu_B}.$$

Substituting the optimal posteriors from Proposition 1 directly yields the corollary. Under Plan A, $\mu_R = \tau_I$ and $\mu_B = 0$ give $q = \mu_0/\tau_I$, $p_r = 1$, and $p_b = \mu_0(1 - \tau_I)/[(1 - \mu_0)\tau_I]$. Under Plan B, $\mu_R = \tau_I$ and $\mu_B = \tau_{II}$ give $q = (\mu_0 - \tau_{II})/(\tau_I - \tau_{II})$, $p_r = \tau_I(\mu_0 - \tau_{II})/[\mu_0(\tau_I - \tau_{II})]$, and $p_b = (1 - \tau_I)(\mu_0 - \tau_{II})/[(1 - \mu_0)(\tau_I - \tau_{II})]$. □

Boundary cases

Proposition 1 assumes $\tau_{II} < \mu_0 < \tau_I$. We now consider the two boundary cases in which the prior lies outside this range.

Prior higher than both thresholds ($\mu_0 \geq \tau_I$). At the prior both types already choose red, so $v(\mu_0) = 1$ is the maximum payoff and no persuasion is needed.

Prior lower than both thresholds ($\mu_0 < \tau_{II}$). The argument parallels Proposition 1. Any feasible weaker posterior has $\mu_B \leq \mu_0 < \tau_{II}$. For an informative plan, $v(\mu_B) = 0$ and $\Pi = q\,v(\mu_R)$. Since $q = (\mu_0 - \mu_B)/(\mu_R - \mu_B)$ is a decreasing function of μ_B, the sender pushes the weaker posterior to $\mu_B = 0$, giving $q = \mu_0/\mu_R$. Given $\mu_B = 0$, the optimal

stronger posterior depends on the share of Type II: setting μ_R to the smallest belief that delivers each value, $\mu_R = \tau_{II}$ gives $\Pi = c_{II}\mu_0/\tau_{II}$ and $\mu_R = \tau_I$ gives $\Pi = \mu_0/\tau_I$. Since $c_{II}\mu_0/\tau_{II} \geq \mu_0/\tau_I \iff c_{II} \geq \tau_{II}/\tau_I$, the sender persuades only Type II ($\mu_R = \tau_{II}$) when $c_{II} \geq \tau_{II}/\tau_I$ and targets Type I ($\mu_R = \tau_I$, Plan A) otherwise, the same threshold as before. Both plans yield a positive payoff, so the optimal plan beats revealing nothing.

B Instructions

B.1 Instructions for the Unknown-High Treatment

B.1.1 Instructions for Part 1

Instructions

Welcome! Thank you for participating in this study of economic decision-making. Your earnings from today's experiment will depend on your decisions, the decisions of others, and luck. If you have questions or need assistance during the experiment, please raise your hand, and an experimenter will assist you. Please turn off your electronic devices and do not talk to other participants during the experiment.

There will be 40 rounds in this experiment. This instruction applies to the first 20 rounds. We will distribute instructions for the last 20 rounds later. Your final earnings will be the sum of your earnings in all 40 rounds plus the $10 show-up bonus.

Overview (for the first 20 rounds)

- There are two different roles in the experiment, **Senders** and **Receivers**.
- Roles will be randomly assigned. Half of you will be **Senders**, and the other half will be **Receivers**.
- All participants will stay in the same role for all 20 rounds.
- However, **Senders** and **Receivers** will be randomly rematched in each round.
- You will make your decisions by choosing a *plan* on the computer, depending on your assigned role:
  - The **Sender's** plan is called a *Communication Plan*.
  - The **Receiver's** plan is called a *Guessing Plan*.

- The computer will then use these plans to determine the outcome of each round.

Sequence of Events in a Single Round

1. The **Receiver** is assigned their type: Type I with a probability of 80% and Type II with a probability of 20%. (The explanation of types will come later)
2. The **Receiver** always knows their type. The **Sender** does not know the **Receiver**'s type.
3. The **Sender** chooses a *Communication Plan*.
4. The **Receiver** chooses a *Guessing Plan*.
5. The computer performs the following procedure based on the **Sender**'s *Communication Plan* and the **Receiver**'s *Guessing Plan*.
   (a) First, the computer randomly draws a ball from a bag that contains 1 red ball and 2 blue balls. (There is a 33.3% probability that the ball will be red and a 66.7% probability that it will be blue.)
   (b) Second, following the **Sender's** *Communication Plan*, a message is sent to the **Receiver** on behalf of the **Sender**. The message can be either "the ball is red" or "the ball is blue."
   (c) Third, following the **Receiver's** *Guessing Plan*, a guess is made on the **Receiver's** behalf about the color of the ball. The guess can be either RED or BLUE.
   (d) Finally, participants will see their earnings for the round.

Earnings of a Round

- The **Sender** earns \$0.4 if the **Receiver's** guess is RED, and \$0 if the **Receiver's** guess is BLUE.

- The **Receiver** earns a payment if their guess is correct (that is, the guess matches the true color of the drawn ball), and \$0 if the guess is incorrect. The amount of the payment depends on the **Receiver's type**:
  - For Type I: The **Receiver** earns \$0.4 for any correct guess, regardless of the color.
  - For Type II: The **Receiver** earns \$0.4 if the guess is RED and correct; The **Receiver** earns \$0.1 if the guess is BLUE and correct;

The Sender's Communication Plan

When the **Sender** chooses their *Communication Plan*, the **Sender** does not know the matched **Receiver's** type.

The purpose of the *Communication Plan* is to tell the computer which message to send when a red ball is drawn and when a blue ball is drawn. It specifies the probability of sending each of the two messages: "the ball is red" and "the ball is blue", under each situation.

Specifically, the **Sender** needs to specify the following two probabilities using slider scales:

1. **If a red ball is drawn, then send the message "the ball is red" with a probability of _____%.**

   By default, the message "the ball is blue" will be sent with the remaining probability, which equals 100 minus the chosen number.

2. **If a blue ball is drawn, then send the message "the ball is red" with a probability of _____%.**

   Again, the message "the ball is blue" will be sent with the remaining probability.

Note that the greater the difference between these two input probabilities, the more informative the *Communication Plan* becomes—that is, the easier it is for **the Receiver** to infer which situation occurred (whether a red ball or blue ball was drawn) from the message.

The Receiver's Guessing Plan

Before the **Receiver** chooses their *Guessing Plan*, they receive the following information:

- The **Receiver's** type (either "Type I" or "Type II") for the current round.
- The **Sender's** *Communication Plan*—that is, the probability with which each message is sent when the drawn ball is red or blue.

The **Receiver** needs to tell the computer how to guess the color of the ball for each possible message they might receive:

1. If the message is "the ball is red", then guess _____(RED/BLUE)
2. If the message is "the ball is blue", then guess _____(RED/BLUE).

Implementation of a Round

After the **Sender** and the **Receiver** submit their *Communication* and *Guessing* Plans, the computer implements the round by executing the following sequence of events:

1. The computer randomly draws a ball from a bag that contains 1 red ball and 2 blue balls.
2. A message is chosen randomly based on the color of the drawn ball and the **Sender's** *Communication Plan*.
    - For example, if a blue ball is drawn and the **Sender's** *Communication Plan* instructs that when a blue ball is drawn, "the ball is red" will be sent with a probability of 40%, then there is a 40% probability that the message will be "the ball is red" and a 60% probability that the message will be "the ball is blue".
3. Based on the message that is sent and the **Receiver's** *Guessing Plan*, the computer enters a guess on behalf of **Receiver**.

- For example, if the message is "the ball is red", and the **Receiver's** *Guessing Plan* instructs the computer to guess RED upon receiving that message, then the final guess will be RED.

4. The computer determines the earnings of the **Sender** and the **Receiver**.

   The **Sender's** earnings depend solely on whether the **Receiver's** guess is RED.

   The **Receiver's** earnings depend on whether the **Receiver's** guess is correct AND on the **Receiver's** type.

   Earnings of a **Sender**:

   - If the **Receiver's** guess is RED, then regardless of the true color of the ball, the **Sender** earns $0.4.
   - If the **Receiver's** guess is BLUE, then regardless of the true color of the ball, the **Sender** earns $0.

   Earnings of a Type I **Receiver**:

   - If the guess is correct, then regardless of the true color of the ball, the **Receiver** earns $0.4.
   - If the guess is incorrect, then regardless of the true color of the ball, the **Receiver** earns $0.

   Earnings of a Type II **Receiver**:

   - If a red ball is drawn AND the **Receiver** correctly guesses RED, the **Receiver** earns $0.4.
   - If a blue ball is drawn AND the **Receiver** correctly guesses BLUE, the **Receiver** earns $0.1.
   - If the guess is incorrect, then regardless of the true color of the ball, the **Receiver** earns $0.

Summary

- There is a 33.3% probability that the ball will be red, and a 66.7% probability that it will be blue.
- There is a 80% probability that the **Receiver** is Type I, and a 20% probability that the **Receiver** is Type II.
- For all 20 rounds, the **Sender** does not know the type of the matched **Receiver**.
- Both the **Sender's** and the **Receiver's** earnings depend on **Receiver's** guess.
- In each round, the **Sender** earns \$0.4 if the **Receiver** guesses RED, and earns \$0 if the **Receiver** guesses BLUE.
- In each round, the **Receiver** earns a payment if their guess is correct, and \$0 if the guess is incorrect. The payment amount depends on the type of the **Receiver**:
  - Type I **Receiver**: earns \$0.4 for any correct guess, regardless of the color.
  - Type II **Receiver**: earns \$0.4 if the guess is RED and is correct; and \$0.1 if the guess is BLUE and correct.
- You will be randomly assigned to one role and will remain in that role for all 20 rounds.
- In each round, you will be randomly paired with another participant.

B.1.2 Instructions for Part 2

Instructions

This instruction applies to the last 20 rounds. Your role remains the same as in the first 20 rounds. However, in the last 20 rounds, there will be no interaction between the **Sender** and the **Receiver**. Each role will perform a separate task independently.

The Sender's Task

In each round, the **Sender**'s task is to guess the *Guessing Plan* of a **Receiver** in a randomly-selected round among the first 20 rounds. Specifically, each round proceeds as follows.

- The computer will randomly pick one round from the first 20 rounds that was played by a different pair of participants. You were not involved in that round and had no influence on its outcome.
- The computer will show you the *Communication Plan* chosen by another **Sender** in that past round, and the type of the matched **Receiver**. Your task is to submit a guess of the **Receiver**'s *Guessing Plan*—that is, to guess how that **Receiver** responded to each possible message.
- After you submit your guess, the computer implements the round as in the first 20 rounds: it draws a ball from the bag (1 red ball and 2 blue balls), sends a message based on the drawn ball and the *Communication Plan*.
- Given the message, your guessed *Guessing Plan* and the **Receiver**'s actual *Guessing Plan* each generate a guess for the ball's color (either RED or BLUE).
- If these two guesses are the same, you earn \$0.4. Otherwise, you earn \$0.

The Receiver's Task

In each round, the **Receiver**'s task is the same as in the first 20 rounds, with one notable difference—your decisions will not affect any other participants in any way. Specifically, each round proceeds as follows.

- The computer will randomly pick one round from the first 20 rounds that was played by a different pair of participants. You were not involved in that round and had no influence on its outcome.
- The computer will show you the *Communication Plan* chosen by the **Sender** in that past round, and the **Receiver**'s type assigned for that round. Your task is to choose a *Guessing Plan* as if you were a **Receiver** of that type.
- After you choose your *Guessing Plan*, the computer implements the round as in the first 20 rounds: it draws a ball from the bag (1 red ball and 2 blue balls), sends a message based on the drawn ball and the *Communication Plan*, and determines your guess based on your *Guessing Plan*.
- Your earnings depend on whether your guess is correct and on the assigned type, as in the first 20 rounds. If the guess is correct, you earn a payment; \$0 if the guess is incorrect. The payment amount depends on the assigned type:
  - Type I **Receiver**: earns \$0.4 for any correct guess, regardless of the color.
  - Type II **Receiver**: earns \$0.4 if the guess is RED and correct; earns \$0.1 if the guess is BLUE and correct.

Summary

- In the last 20 rounds, there is no interaction between the **Sender** and the **Receiver**.
- The **Sender**'s task is to guess the *Guessing Plan* of a **Receiver** in a randomly-selected past round. The **Sender** earns \$0.4 if the guess matches the **Receiver**'s guess, and \$0 otherwise.
- The **Receiver**'s task is to choose a *Guessing Plan* given the *Communication Plan* and **Receiver** type from a randomly-selected past round. The **Receiver** earns a positive amount if the guess is correct, and \$0 if incorrect. The payment amount depends on the **Receiver**'s type:
  - Type I **Receiver**: earns \$0.4 for any correct guess, regardless of the color.
  - Type II **Receiver**: earns \$0.4 if the guess is RED and correct; and \$0.1 if the guess is BLUE and correct.
- Unlike the first 20 rounds, your decisions in the last 20 rounds affect only your own payoffs and have no effect on other participants.

B.2 Instructions for the Unknown-Low Treatment

B.2.1 Instructions for Part 1

Instructions

Welcome! Thank you for participating in this study of economic decision-making. Your earnings from today's experiment will depend on your decisions, the decisions of others, and luck. If you have questions or need assistance during the experiment, please raise your hand, and an experimenter will assist you. Please turn off your electronic devices and do not talk to other participants during the experiment.

There will be 40 rounds in this experiment. This instruction applies to the first 20 rounds. We will distribute instructions for the last 20 rounds later. Your final earnings will be the sum of your earnings in all 40 rounds plus the $10 show-up bonus.

Overview (for the first 20 rounds)

- There are two different roles in the experiment, **Senders** and **Receivers**.
- Roles will be randomly assigned. Half of you will be **Senders**, and the other half will be **Receivers**.
- All participants will stay in the same role for all 20 rounds.
- However, **Senders** and **Receivers** will be randomly rematched in each round.
- You will make your decisions by choosing a *plan* on the computer, depending on your assigned role:
  - The **Sender's** plan is called a *Communication Plan*.
  - The **Receiver's** plan is called a *Guessing Plan*.
- The computer will then use these plans to determine the outcome of each round.

Sequence of Events in a Single Round

1. The **Receiver** is assigned their type: Type I with a probability of 20% and Type II with a probability of 80%. (The explanation of types will come later)
2. The **Receiver** always knows their type. The **Sender** does not know the **Receiver**'s type.
3. The **Sender** chooses a *Communication Plan*.
4. The **Receiver** chooses a *Guessing Plan*.
5. The computer performs the following procedure based on the **Sender**'s *Communication Plan* and the **Receiver**'s *Guessing Plan*.
   (a) First, the computer randomly draws a ball from a bag that contains 1 red ball and 2 blue balls. (There is a 33.3% probability that the ball will be red and a 66.7% probability that it will be blue.)
   (b) Second, following the **Sender's** *Communication Plan*, a message is sent to the **Receiver** on behalf of the **Sender**. The message can be either "the ball is red" or "the ball is blue."
   (c) Third, following the **Receiver's** *Guessing Plan*, a guess is made on the **Receiver's** behalf about the color of the ball. The guess can be either RED or BLUE.
   (d) Finally, participants will see their earnings for the round.

Earnings of a Round

- The **Sender** earns \$0.4 if the **Receiver's** guess is RED, and \$0 if the **Receiver's** guess is BLUE.

- The **Receiver** earns a payment if their guess is correct (that is, the guess matches the true color of the drawn ball), and \$0 if the guess is incorrect. The amount of the payment depends on the **Receiver's type**:
  - For Type I: The **Receiver** earns \$0.4 for any correct guess, regardless of the color.
  - For Type II: The **Receiver** earns \$0.4 if the guess is RED and correct; The **Receiver** earns \$0.1 if the guess is BLUE and correct;

The Sender's Communication Plan

When the **Sender** chooses their *Communication Plan*, the **Sender** does not know the matched **Receiver's** type.

The purpose of the *Communication Plan* is to tell the computer which message to send when a red ball is drawn and when a blue ball is drawn. It specifies the probability of sending each of the two messages: "the ball is red" and "the ball is blue", under each situation.

Specifically, the **Sender** needs to specify the following two probabilities using slider scales:

1. **If a red ball is drawn, then send the message "the ball is red" with a probability of _____%.**

   By default, the message "the ball is blue" will be sent with the remaining probability, which equals 100 minus the chosen number.

2. **If a blue ball is drawn, then send the message "the ball is red" with a probability of _____%.**

   Again, the message "the ball is blue" will be sent with the remaining probability.

Note that the greater the difference between these two input probabilities, the more informative the *Communication Plan* becomes—that is, the easier it is for **the Receiver** to infer which situation occurred (whether a red ball or blue ball was drawn) from the message.

The Receiver's Guessing Plan

Before the **Receiver** chooses their *Guessing Plan*, they receive the following information:

- The **Receiver's** type (either "Type I" or "Type II") for the current round.
- The **Sender's** *Communication Plan*—that is, the probability with which each message is sent when the drawn ball is red or blue.

The **Receiver** needs to tell the computer how to guess the color of the ball for each possible message they might receive:

1. If the message is "the ball is red", then guess _____(RED/BLUE)
2. If the message is "the ball is blue", then guess _____(RED/BLUE).

Implementation of a Round

After the **Sender** and the **Receiver** submit their *Communication* and *Guessing* Plans, the computer implements the round by executing the following sequence of events:

1. The computer randomly draws a ball from a bag that contains 1 red ball and 2 blue balls.
2. A message is chosen randomly based on the color of the drawn ball and the **Sender's** *Communication Plan*.
    - For example, if a blue ball is drawn and the **Sender's** *Communication Plan* instructs that when a blue ball is drawn, "the ball is red" will be sent with a probability of 40%, then there is a 40% probability that the message will be "the ball is red" and a 60% probability that the message will be "the ball is blue".
3. Based on the message that is sent and the **Receiver's** *Guessing Plan*, the computer enters a guess on behalf of **Receiver**.

- For example, if the message is "the ball is red", and the **Receiver's** *Guessing Plan* instructs the computer to guess RED upon receiving that message, then the final guess will be RED.

4. The computer determines the earnings of the **Sender** and the **Receiver**.

   The **Sender's** earnings depend solely on whether the **Receiver's** guess is RED.

   The **Receiver's** earnings depend on whether the **Receiver's** guess is correct AND on the **Receiver's** type.

   Earnings of a **Sender**:

   - If the **Receiver's** guess is RED, then regardless of the true color of the ball, the **Sender** earns $0.4.
   - If the **Receiver's** guess is BLUE, then regardless of the true color of the ball, the **Sender** earns $0.

   Earnings of a Type I **Receiver**:

   - If the guess is correct, then regardless of the true color of the ball, the **Receiver** earns $0.4.
   - If the guess is incorrect, then regardless of the true color of the ball, the **Receiver** earns $0.

   Earnings of a Type II **Receiver**:

   - If a red ball is drawn AND the **Receiver** correctly guesses RED, the **Receiver** earns $0.4.
   - If a blue ball is drawn AND the **Receiver** correctly guesses BLUE, the **Receiver** earns $0.1.
   - If the guess is incorrect, then regardless of the true color of the ball, the **Receiver** earns $0.

Summary

- There is a 33.3% probability that the ball will be red, and a 66.7% probability that it will be blue.
- There is a 20% probability that the **Receiver** is Type I, and a 80% probability that the **Receiver** is Type II.
- For all 20 rounds, the **Sender** does not know the type of the matched **Receiver**.
- Both the **Sender's** and the **Receiver's** earnings depend on **Receiver's** guess.
- In each round, the **Sender** earns \$0.4 if the **Receiver** guesses RED, and earns \$0 if the **Receiver** guesses BLUE.
- In each round, the **Receiver** earns a payment if their guess is correct, and \$0 if the guess is incorrect. The payment amount depends on the type of the **Receiver**:
  - Type I **Receiver**: earns \$0.4 for any correct guess, regardless of the color.
  - Type II **Receiver**: earns \$0.4 if the guess is RED and is correct; and \$0.1 if the guess is BLUE and correct.
- You will be randomly assigned to one role and will remain in that role for all 20 rounds.
- In each round, you will be randomly paired with another participant.

B.2.2 Instructions for Part 2

Instructions

This instruction applies to the last 20 rounds. Your role remains the same as in the first 20 rounds. However, in the last 20 rounds, there will be no interaction between the **Sender** and the **Receiver**. Each role will perform a separate task independently.

The Sender's Task

In each round, the **Sender**'s task is to guess the *Guessing Plan* of a **Receiver** in a randomly-selected round among the first 20 rounds. Specifically, each round proceeds as follows.

- The computer will randomly pick one round from the first 20 rounds that was played by a different pair of participants. You were not involved in that round and had no influence on its outcome.
- The computer will show you the *Communication Plan* chosen by another **Sender** in that past round, and the type of the matched **Receiver**. Your task is to submit a guess of the **Receiver**'s *Guessing Plan*—that is, to guess how that **Receiver** responded to each possible message.
- After you submit your guess, the computer implements the round as in the first 20 rounds: it draws a ball from the bag (1 red ball and 2 blue balls), sends a message based on the drawn ball and the *Communication Plan*.
- Given the message, your guessed *Guessing Plan* and the **Receiver**'s actual *Guessing Plan* each generate a guess for the ball's color (either RED or BLUE).
- If these two guesses are the same, you earn \$0.4. Otherwise, you earn \$0.

The Receiver's Task

In each round, the **Receiver**'s task is the same as in the first 20 rounds, with one notable difference—your decisions will not affect any other participants in any way. Specifically, each round proceeds as follows.

- The computer will randomly pick one round from the first 20 rounds that was played by a different pair of participants. You were not involved in that round and had no influence on its outcome.
- The computer will show you the *Communication Plan* chosen by the **Sender** in that past round, and the **Receiver**'s type assigned for that round. Your task is to choose a *Guessing Plan* as if you were a **Receiver** of that type.
- After you choose your *Guessing Plan*, the computer implements the round as in the first 20 rounds: it draws a ball from the bag (1 red ball and 2 blue balls), sends a message based on the drawn ball and the *Communication Plan*, and determines your guess based on your *Guessing Plan*.
- Your earnings depend on whether your guess is correct and on the assigned type, as in the first 20 rounds. If the guess is correct, you earn a payment; \$0 if the guess is incorrect. The payment amount depends on the assigned type:
  - Type I **Receiver**: earns \$0.4 for any correct guess, regardless of the color.
  - Type II **Receiver**: earns \$0.4 if the guess is RED and correct; earns \$0.1 if the guess is BLUE and correct.

Summary

- In the last 20 rounds, there is no interaction between the **Sender** and the **Receiver**.
- The **Sender**'s task is to guess the *Guessing Plan* of a **Receiver** in a randomly-selected past round. The **Sender** earns \$0.4 if the guess matches the **Receiver**'s guess, and \$0 otherwise.
- The **Receiver**'s task is to choose a *Guessing Plan* given the *Communication Plan* and **Receiver** type from a randomly-selected past round. The **Receiver** earns a positive amount if the guess is correct, and \$0 if incorrect. The payment amount depends on the **Receiver**'s type:
  - Type I **Receiver**: earns \$0.4 for any correct guess, regardless of the color.
  - Type II **Receiver**: earns \$0.4 if the guess is RED and correct; and \$0.1 if the guess is BLUE and correct.
- Unlike the first 20 rounds, your decisions in the last 20 rounds affect only your own payoffs and have no effect on other participants.

C Sample Interfaces

Receiver's Payoff in that round (Type-I Receiver)		
Guess RED and correct	Guess BLUE and correct	Guess wrong
$0.40	$0.40	$0

Your payment for this round	
If your guess matches the Receiver's guess	If it does not match
$0.40	$0

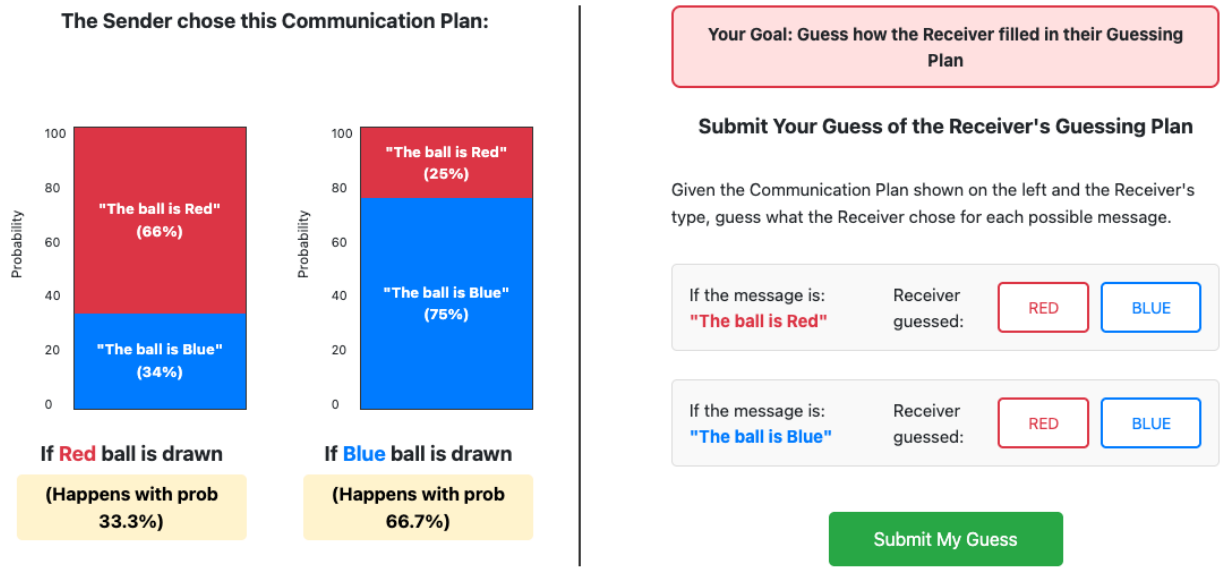


Figure A.1: Sample interface of a sender's decision in Part 2.

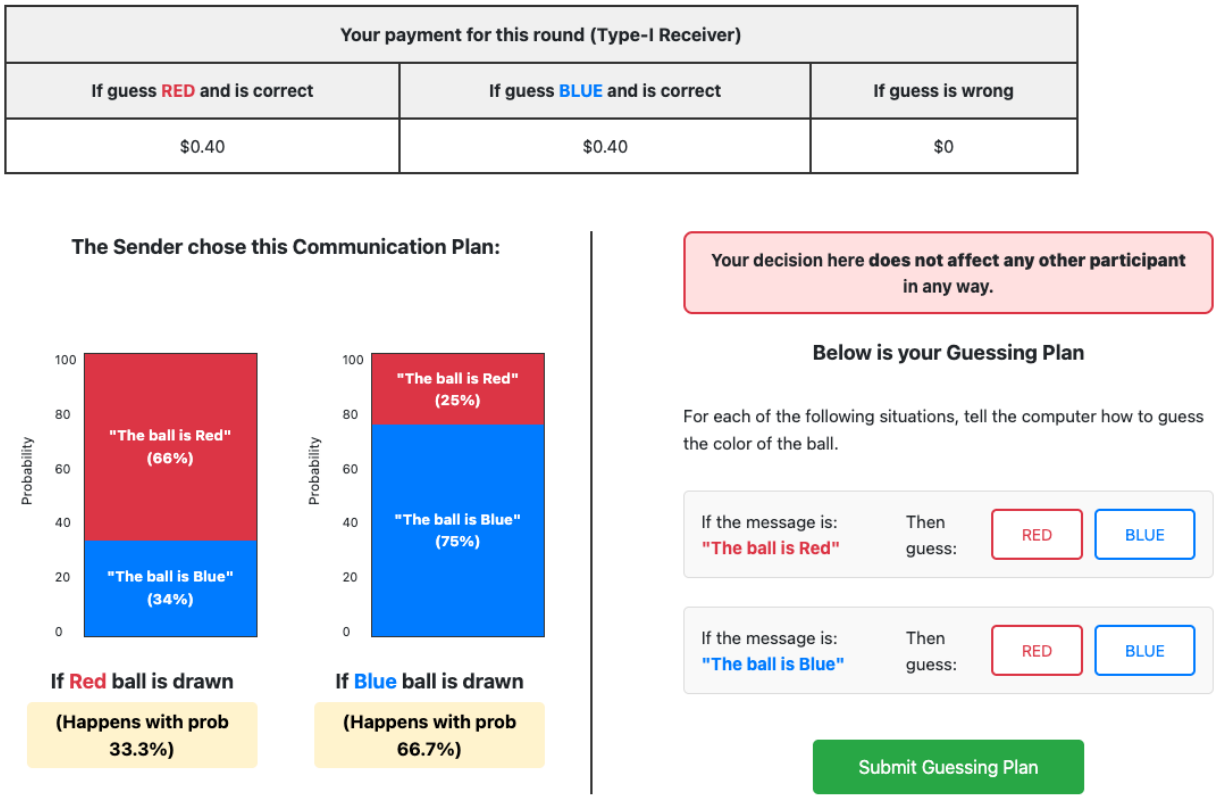


Figure A.2: Sample interface of a receiver's decision in Part 2.

We would like to ask you about the probability that something will happen. We would like you to give a number from 0 to 100, where "0" means you think it will definitely not happen, and "100" means you think it will definitely happen.

Consider a bowl with 10 balls in total. Some of the balls may be white and some red.

Question 1

First, suppose this bowl has 10 white balls and no red balls. You will be asked to draw one ball without looking. On a scale from 0 percent to 100 percent, what is the probability that the ball you draw is **red**?

out of 100

Question 2

Now suppose that the bowl has 7 white balls and 3 red balls. You will be asked to draw one ball without looking. What is the probability that the ball you draw is **white**?

out of 100

Question 3

Imagine that the weather report tells you that the probability it will rain tomorrow is 70%. Assume the weather report accurately reports the probability of rain, what is the probability it will NOT rain tomorrow?

out of 100

Question 4

Imagine that whether it rains in College Station and whether it rains in Paris are unrelated. The probability that it will rain in College Station tomorrow is 50%. The probability that it will rain in Paris is also 50%. What is the probability that it will rain both in College Station and in Paris tomorrow?

out of 100

Submit

Figure A.3: Probability numeracy questions in the post-experiment survey.